Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

# SDC Hackathon

## Phase 1 : Preliminary Project Proposal

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

# 1. TEAM AND PROJECT OVERVIEW

## Team Name

Nexgen

## Selected Domain

Domain 2 - Orbyte Smart SME

## Selected SDG Track (1–17)

SDG 8 (main); SDG 12 (related)

## Project / Solution Name

NexFlow

## Name (Leader)

TODO - HUMAN VERIFICATION - registered leader

## Email & Phone (Leader)

TODO - HUMAN VERIFICATION - email and phone

## University Affiliation(s)

TODO - HUMAN VERIFICATION - institution(s)

## Number of Team Members

TODO - HUMAN VERIFICATION - registered member count

# 2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

## 2.1. The Problem Context

*Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.*

***[Maximum 250 words]***

Independent bakeries must count unsold bread, decide clearance prices, write promotions and arrange the remaining food at closing. These repeated tasks fragment the daily record and take staff attention away from serving customers. Owners need a clear approval step and visibility of what was paid, collected or handed over.

NexFlow addresses this single-store operational problem. No store interviews, commercial results or measured time savings are claimed in this submission. The proposed value is improved consistency and productivity for small businesses (SDG 8), with responsible surplus handling as a related SDG 12 direction.

# 2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

## 2.2. The Proposed AI Solution

*Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.*

***[Maximum 300 words]***

NexFlow provides a three-language bakery closing workspace: staff enter surplus quantities; deterministic code calculates prices and food-footprint estimates; optional DeepSeek drafts promotional wording; the owner reviews it; customers simulate buying and collecting surprise boxes; staff record remaining-food handover. The dashboard reads the same backend ledger.

The default competition runner uses a local copy template, synthetic inventory, simulated payment and WhatsApp messages. A separate YOLOv8 prototype analyzes still images with staff correction; it does not update the main inventory. The animated welcome-page bread count is a preset replay, not live inference.

Workshop-style technologies applied include API-based text generation, SQLite records and cloud hosting on Replit. Individual workshop attendance requires team confirmation.

# 3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

## 3.1. Conceptual Tech Stack

*List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.*

*Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.*

Main app: Python, FastAPI, Pydantic, Starlette, Uvicorn, SQLite; native HTML/CSS/JavaScript and three-language dictionaries. HTTPX supports the optional DeepSeek API. Browser -> FastAPI -> SQLite/catalogue; optional API copy returns to owner review.

Independent vision prototype: Ultralytics YOLOv8, OpenCV, NumPy and PyTorch. Browser ONNX experiments remain separate. Replit hosts a public project deployment; some assets differ from this GitHub baseline.

Live prototype: https://orbyte-smart-yield-nexgen.replit.app/
RunPod: Not used. Replit sponsor-credit redemption, plan and region: TODO - HUMAN VERIFICATION.

# 3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

## 3.2. Design-Thinking & Deployment Strategy

*Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.*

***[Maximum 200 words]***

The workflow centers on one repeated task: closing a bakery. Staff enter quantities, owners approve offers, and customers see a focused purchase page. English, Bahasa Melayu and Chinese support the demonstration. Humans confirm promotions, counts and handovers.

The prototype separates calculated facts from generated wording and simulated external actions. Automated tests cover business rules; a fresh temporary session supports a repeatable demo. Replit provides a publicly reachable showcase and workspace, while the local runner is the reproducible baseline.

A future pilot should compare closing-task time and count corrections against the same manual scenario. This is a proposed evaluation, not evidence of completed user research or deployment readiness.

# 4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

*To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.*

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

# 4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

| Metric Category | Your Response |
|---|---|
| **Target SDG Sub-Target(s)**<br>*Identify the exact UN SDG sub-target your project addresses*<br><br>*e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."* | Main: SDG 8.2 (productivity through technology) and 8.3 (small-enterprise growth). Related: SDG 12.3 (retail/consumer food waste).<br>Sources: sdgs.un.org/goals/goal8 and sdgs.un.org/goals/goal12. |
| **Current Problem Baseline**<br>*What is the starting condition of the problem, expressed as a number?*<br><br>*e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."* | Synthetic baseline: 10 croissants, retail RM60, cost RM20, 0.8 kg food; demo clearance forecast RM30 and food-footprint estimate 2 kg CO2e. These are catalogue/code fixtures, not a surveyed store or verified avoided emissions.<br>Real operational baseline: not measured; human pilot needed. |

# 4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

| Metric Category | Your Response |
| --- | --- |
| **Core Measurable KPI**<br>*The specific unit you will track*<br><br>*e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided* | Track minutes per closing task, count corrections, food items picked up or handed over, and simulated amount paid. Do not equate the dashboard footprint estimate with certified emissions savings. |
| **Phase 2 Impact Target**<br>*Your realistic, percentage-based target by the final prototype presentation*<br><br>*e.g., "20% improvement in computational resource use."* | TODO - HUMAN VERIFICATION: no approved percentage target or measured comparison exists. Suggested pilot method: repeat a fixed manual scenario with and without NexFlow, then agree a target with the team. No invented improvement percentage is submitted. |

# 5. ELIGIBILITY & READINESS CHECKLIST

*Tick all boxes that apply before submitting.*

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

HUMAN ACTION: verify workshop attendance and all checklist statements. No compliance boxes have been checked on behalf of the team.

# 6. IP & ORIGINALITY DECLARATION

## 6.1. Pre-Existing / Prior-Owned Work

*Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?*

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

TODO - HUMAN VERIFICATION: each member must confirm whether any prior-owned code, design, model or research predates the event. Do not assume No from Git history.

# 6. IP & ORIGINALITY DECLARATION

## 6.2. Third-Party Code, Datasets, or Models

*Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?*

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

| Name | Type (code/dataset/model) | License |
|---|---|---|
| FastAPI / Pydantic / HTTPX | Code; others in declaration | MIT / BSD / Apache; see declaration |
| YOLOv8 / Bread Detector | Model / dataset; see declaration | AGPL-3.0 / CC BY 4.0; see appendix |
| ONNX Runtime Web 1.17.3 | Browser model runtime | MIT |
| Tailwind / fonts / AI tools | Code / content; see declaration | See technology declaration |
| Additional dependencies | Complete details in declaration appendix | See technology declaration |

# 6. IP & ORIGINALITY DECLARATION

## 6.3. Employment-Related IP Conflicts

*Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?*

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

TODO - HUMAN VERIFICATION: all members must confirm employment, internship and consulting IP obligations. No Yes/No selection has been made on their behalf.

# 7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]

Signature and date deliberately left for the authorized team leader. This document contains unresolved human-verification fields and is not ready for final upload.